

**Notes: Azevedo and Leshno (JPE, 2016)**  
**A Supply and Demand Framework for Two-Sided Matching Markets**

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# 1 One-sentence summary

Azevedo and Leshno build a continuum model of many-student, finite-college matching markets in which stable matchings are characterized by supply-demand equations and college cutoffs play the role of prices.

## 2 Big picture

### 2.1 Main question

How can economists obtain price-theoretic comparative statics for two-sided matching markets when preferences are heterogeneous and transfers are absent or limited?

### 2.2 Central idea

In large school-choice or college-admissions markets, a finite set of colleges faces a continuum of students. A stable matching can be summarized by college-specific cutoffs. These cutoffs operate like prices: they determine which students demand which colleges, and market clearing equates demand to capacity.

### 2.3 Why the paper matters

The Gale-Shapley framework is excellent for existence, stability, and mechanism design, but it is often hard to get clean comparative statics. By turning stable matching into a cutoff-equilibrium model, the paper makes matching markets look like supply and demand systems.

### 2.4 Main result in plain language

Under regular conditions, the continuum economy has a unique stable matching. Moreover, large finite Gale-Shapley markets converge to the continuum matching, and the set of stable matchings in finite markets becomes small.

### 2.5 Applications

The framework is used to study school competition, convergence rates of large matching markets, and school choice mechanisms such as random serial dictatorship.

## 3 Incremental contribution of the paper

### 3.1 A price-theoretic representation of stable matching

The paper shows that stable matchings in a large matching market can be represented by a vector of cutoffs, and those cutoffs clear markets just like prices in standard supply-and-demand models.

### 3.2 Bridges Gale-Shapley and supply-demand analysis

The paper combines the flexibility of Gale-Shapley matching with the comparative-statics logic of price theory. This is the core conceptual contribution.

### 3.3 Continuum approximation for large discrete markets

The paper proves that, when the continuum economy has a unique stable matching, stable matchings of large finite economies converge to the continuum cutoffs. This justifies using the simpler continuum model to study large school-choice systems.

### 3.4 Generic uniqueness

Stable matchings can be multiple in finite matching markets. The paper shows that in the continuum model uniqueness is typical: full support guarantees uniqueness, and regular economies have unique stable matchings for almost all capacity vectors.

### 3.5 Comparative statics for school quality and mechanisms

The model makes it possible to ask quantitative comparative-statics questions: Which school-quality investments attract better students? How fast do finite markets converge? How do mechanisms such as random serial dictatorship behave in the limit?

## 4 Conceptual framework

### 4.1 Agents and types

There is a finite set of colleges, denoted by  $c = 1, \dots, C$ , and a continuum of students. A student type contains preferences over colleges and scores at each college. The set of types is denoted by  $\Theta$ .

### 4.2 College capacities

Each college  $c$  has capacity  $S_c$ . The model permits total capacity to be less than one, so some students may remain unmatched.

### 4.3 Cutoffs as prices

A cutoff vector is

$$P = (P_1, \dots, P_C).$$

Student  $\theta$  is eligible for college  $c$  if her score at college  $c$  exceeds  $P_c$ . Given cutoffs, each student chooses her most preferred feasible college.

### 4.4 Demand for colleges

Let  $D_c(P)$  be the measure of students who choose college  $c$  at cutoff vector  $P$ . Demand depends on preferences, scores, and the cutoff vector. Higher  $P_c$  lowers demand for college  $c$ , just as a higher price lowers demand in a standard market.

### 4.5 Market clearing

A cutoff vector  $P$  clears the market if

$$D_c(P) \leq S_c \quad \text{for all } c,$$

with equality for colleges whose cutoffs are positive:

$$P_c > 0 \quad \Rightarrow \quad D_c(P) = S_c.$$

## 4.6 Stable matching

A stable matching is an assignment with no blocking student-college pair. The supply and demand lemma shows that stable matchings and market-clearing cutoff vectors are equivalent.

# 5 Model setup and derivations

## 5.1 Supply and demand lemma

**Lemma 1.** If  $m$  is a stable matching, then there exists a market-clearing cutoff vector  $P$  such that  $m = M_P$ . Conversely, if  $P$  is a market-clearing cutoff vector, then  $M_P$  is a stable matching.

**Detailed interpretation:** The lemma is the paper’s fundamental translation result. It turns stability into market clearing. Once this is done, comparative statics can be done by studying how demand curves shift and how cutoffs change.

## 5.2 Demand mechanics

Given cutoffs, a student considers the feasible colleges:

$$F(\theta, P) = \{c : e_c(\theta) \geq P_c\}.$$

The student chooses her favorite college in  $F(\theta, P)$ , or remains unmatched if the set is empty. Aggregate demand is the measure of types choosing each college.

## 5.3 Regularity and uniqueness

Regularity rules out knife-edge economies in which demand is flat exactly at market-clearing capacity over a large region. If the distribution of types has full support, demand is sufficiently responsive to cutoffs to yield a unique stable matching.

## 5.4 Theorem 1: uniqueness

**Theorem 1.** If the distribution of student types has full support, then the continuum economy has a unique stable matching. More generally, if the distribution is regular, then for almost every vector of capacities the economy has a unique stable matching.

**Detailed interpretation:** This is a major result because classical finite matching markets can have many stable matchings. The theorem says that multiplicity is not the typical case in large continuum environments.

## 5.5 Theorem 2: convergence of finite economies

**Theorem 2.** If a continuum economy has a unique stable matching, then any sequence of stable matchings of finite economies converging to it converges to the continuum stable matching. The diameter of the set of finite stable matchings goes to zero, and the stable matching correspondence is continuous at the continuum economy.

**Detailed interpretation:** This theorem justifies the continuum approximation. It says that, for large markets, the continuum cutoff vector is not merely convenient; it is the limiting description of the discrete Gale-Shapley outcomes.

## 5.6 Nonrobustness of multiple stable matchings

**Proposition 1.** If a continuum economy has multiple stable matchings ordered around a cutoff vector, then nearby perturbations of fundamentals can remove stable matchings in any small neighborhood of that cutoff.

**Detailed interpretation:** Multiple stable matchings are fragile. They can exist, but small changes in capacity or type distributions may eliminate them. This reinforces the use of generic uniqueness.

## 6 Applications in the paper

### 6.1 School quality and competition

The paper studies how an increase in school quality  $d_c$  affects the quality of the school's entering class:

$$Q_c(d) = \int_{m_d(c)} e_c(\theta) dh_d(\theta).$$

The effect can be decomposed into the characteristics of marginal students who switch schools or cross cutoff boundaries.

### 6.2 Interpreting marginal students

The impact of improving a school's quality depends on which students are induced to choose it and which students are displaced by capacity constraints. This mirrors the way price theory evaluates demand shifts by looking at marginal consumers.

### 6.3 Convergence rates

The paper simulates finite markets approximating the continuum example. It studies how the fraction of students receiving different matches and the cutoff vector converge as market size grows.

### 6.4 Random serial dictatorship

The paper applies the framework to random serial dictatorship and shows that in large markets RSD can be represented by a supply-demand system with a scalar parameter related to how many students get priority positions.

## 7 Identification and interpretation

### 7.1 What is identified theoretically

This is a theory paper. The identification-style question is not causal estimation but model interpretation: what stable matching object is equivalent to a price system, and when does the continuum object describe finite markets?

### 7.2 Why the continuum limit is useful

Large matching markets often have many participants on one side and a relatively small number of institutions on the other. The continuum assumption abstracts from individual student identities while preserving heterogeneity in types. This produces smooth demand functions and price-like cutoffs.

### 7.3 Main assumptions

- A continuum of students and a finite set of colleges.
- Preferences and scores are summarized in student types.
- Colleges use cutoffs over scores.
- Stability is the equilibrium notion.
- Regularity or full support ensures uniqueness.

### 7.4 What the model does not do

The model does not estimate behavior from data. It does not model strategic misreporting by students, endogenous priorities, or transfers. It is a theoretical framework that makes stable matching analytically tractable through market-clearing cutoffs.

### 7.5 Policy interpretation

The paper helps market designers use simulations and comparative statics with confidence in large markets. If the continuum economy is unique, small errors in fundamentals or finite-market sampling noise should not radically change stable outcomes.

## 8 Tables and figures

### 8.1 Figure 1: The Gale and Shapley algorithm

**Read:**

- The figure illustrates the student-proposing Gale-Shapley algorithm in a two-college example.
- The squares represent student type spaces, and rows show successive steps of the algorithm.
- Dark gray and light gray regions are tentatively assigned to colleges 1 and 2.
- White regions are not yet assigned or are unmatched.

**Economic message:** The algorithmic matching process can be summarized by cutoffs. The figure motivates the move from iterative matching to cutoff-based market clearing.

**Detailed interpretation:** Figure 1 shows how the Gale-Shapley algorithm gradually induces threshold-like regions in type space. The continuum model abstracts from the iteration and directly studies the limiting cutoff vector.

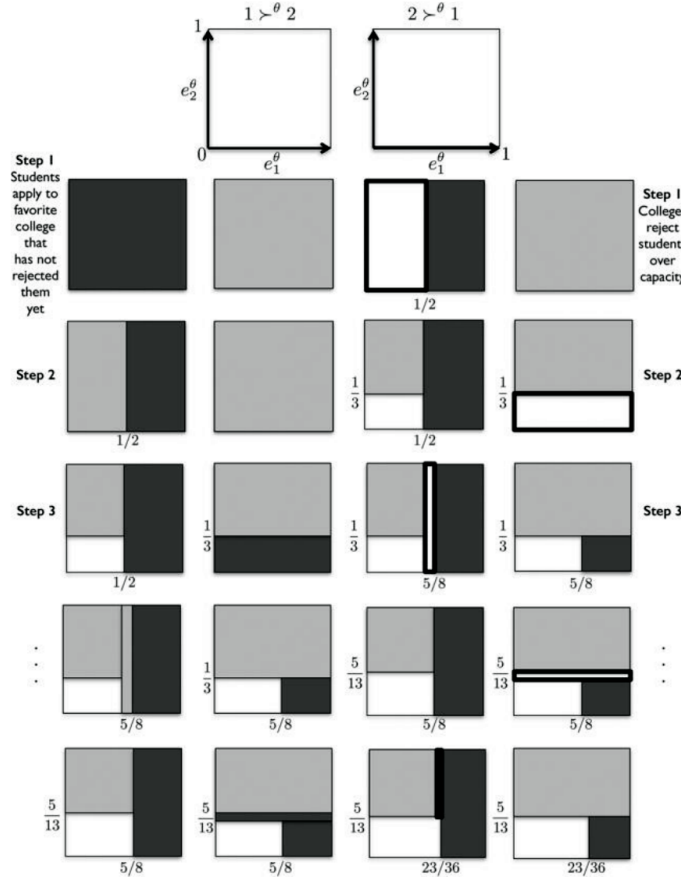


FIG. 1.—The Gale and Shapley algorithm. The set of student types  $\Theta$  is represented by the two squares on the top panel. The left square represents students who prefer college 1 and the right square represents students who prefer college 2. Scores at each college are represented by the  $(x, y)$  coordinates. The lower panels show the first five steps of the Gale-Shapley student-proposing algorithm. In each line, students apply to their favorite colleges that have not yet rejected them in the left panel, and colleges reject students to remain within capacity in the right panel. Students in dark gray are tentatively assigned to college 1 and in light gray

**Figure 1: The Gale and Shapley algorithm.** Original figure cropped from the paper. It visually motivates why stable assignments in a large market can be represented by cutoff boundaries.

## 8.2 Figure 2: A stable matching in a continuum economy; Figure 3: A stable matching in a discrete economy

**Read:**

- Figure 2 shows the cutoff representation in a continuum economy with two colleges.
- Figure 3 shows a discrete economy approximating the same continuum environment.
- Circles, squares, and X marks represent students assigned to college 1, college 2, or unmatched.
- The discrete assignment visually approximates the continuum cutoff regions.

**Economic message:** These figures summarize the paper’s approximation argument: large discrete matching markets behave like continuum cutoff economies.



**Detailed interpretation:** Figure 2 is the idealized supply-demand object. Figure 3 is the finite-market realization. The similarity between them previews Theorem 2, which formalizes convergence.

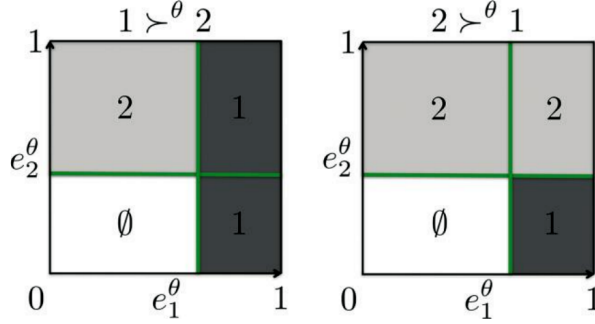


FIG. 2.—A stable matching in a continuum economy with two colleges. The two square represent the set of student types  $\Theta$ . The left square represents students who prefer college 1 and the right square students who prefer college 2. Scores at each college are represented by the  $(x, y)$  coordinates. The white area represents unmatched students, dark

(a) **Figure 2: Continuum stable matching.** Cutoff lines partition the type space into college-1, college-2, and unmatched regions.

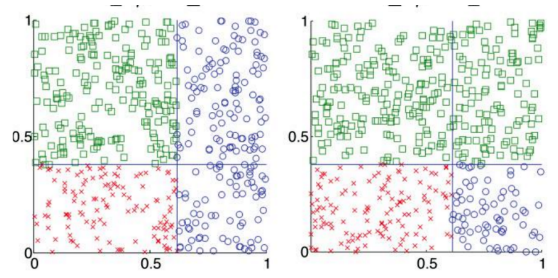


FIG. 3.—A stable matching in a discrete economy approximating the example. The two squares represent the set of student types  $\Theta$ . The left square represents students who prefer college 1 and the right square students who prefer college 2. Scores at each college are represented by the  $(x, y)$  coordinates. There are two colleges, with capacities 250 and 500: 500 students have preferences  $\succ^{\theta} = 1, 2, \emptyset$  and 500 students have preferences  $2, 1, \emptyset$ . Scores  $e^{\theta}$  were drawn independently according to the uniform distribution in  $[0, 1]^2$ . The figure depicts the student-optimal stable matching. Circles represent students matched to college 1, squares

(b) **Figure 3: Discrete approximation.** A finite economy realizes the same cutoff logic with individual student points.

**Figures 2 and 3.** These adjacent figures are placed on the same row. Together, they show the core continuum approximation: cutoffs in the continuum model describe the large finite matching market.

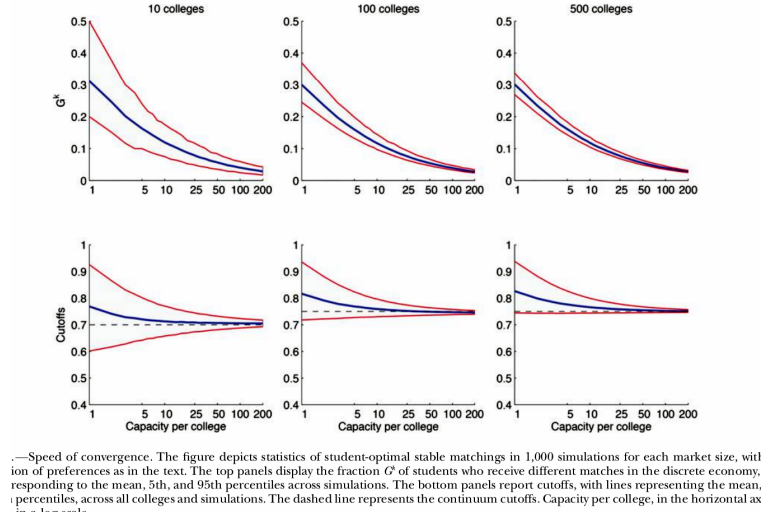
### 8.3 Figure 4: Speed of convergence

**Read:**

- The top panels report the fraction of students receiving different matches in finite simulations.
- The bottom panels report the convergence of cutoffs.
- Columns vary the number of colleges: 10, 100, and 500.
- The horizontal axis is capacity per college, and convergence improves as market size grows.

**Economic message:** The figure quantifies when the continuum model is a good approximation. Larger markets and larger capacities make finite stable matchings close to continuum cutoffs.

**Detailed interpretation:** Figure 4 is an empirical-style simulation supporting the asymptotic theory. It shows that the continuum approximation is not just elegant; it can be accurate in reasonably large markets.



**Figure 4: Speed of convergence.** Original figure cropped and rotated into readable orientation. The figure reports simulation evidence on how quickly finite matchings converge to the continuum stable matching.

## 8.4 Notes-created table: theoretical results and interpretations

| Result                  | Formal content                                                                   | Interpretation                                                      |
|-------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------|
| Supply and demand lemma | Stable matchings correspond to market-clearing cutoff vectors.                   | Cutoffs play the role of prices in matching markets.                |
| Theorem 1               | Full support or generic regularity implies unique continuum stable matching.     | Multiplicity is not typical in the continuum model.                 |
| Theorem 2               | Finite stable matchings converge to continuum stable matching when it is unique. | The continuum model approximates large Gale-Shapley markets.        |
| Proposition 1           | Multiple stable matchings are nonrobust to small perturbations.                  | Multiplicity is fragile and less useful for comparative statics.    |
| Proposition 2           | School quality improvements can be evaluated through marginal students.          | Comparative statics resemble price-theoretic demand-shift analysis. |
| Propositions 3–5        | Large-market mechanisms such as RSD have cutoff limits.                          | Market-design mechanisms can be studied with supply-demand tools.   |

**Notes-created table: theoretical results.** The original paper has no empirical tables. This compact table summarizes the main formal results and the role each result plays in the paper.

## 9 Short summary

- The paper provides a supply-and-demand framework for two-sided matching markets.
- Stable matchings are characterized by market-clearing cutoffs.
- College cutoffs play the role of prices.
- Continuum economies typically have a unique stable matching.
- Large finite Gale-Shapley economies converge to the continuum cutoff equilibrium.
- Multiple stable matchings are nonrobust to small perturbations.

- The framework enables comparative statics for school quality and market design mechanisms.
- The paper bridges market design and price theory.

## 10 Conclusion

### 10.1 Core conclusion

Azevedo and Leshno show that stable matching can be analyzed as a supply-and-demand equilibrium when one side of the market is large. College cutoffs operate like prices, and stable matchings correspond to cutoffs that clear markets.

### 10.2 Economic intuition

In a matching market, a student cannot simply demand her favorite college. She must also be accepted. A college cutoff summarizes the acceptance constraint. Given cutoffs, students choose their favorite feasible school. Aggregate student choices generate demand, and cutoffs adjust until demand equals capacity.

### 10.3 Incremental contribution revisited

The paper's main contribution is to make stable matching comparative-statics-friendly. It takes the complex discrete Gale-Shapley model and shows that, in large markets, the relevant object is a unique cutoff vector that responds continuously to fundamentals.

### 10.4 Implications for market design

The framework supports the use of simulations and policy analysis in large school-choice markets. If the continuum economy is unique, then finite-market outcomes should be stable under small perturbations and approximately independent of which finite stable matching is selected.

### 10.5 Limitations

The continuum framework is an approximation. It is most convincing in large markets with many students and a fixed number of colleges. Small markets, strategic reporting, institutional priority rules, and nonregular distributions may require more detailed discrete analysis.

### 10.6 Takeaway

The paper's lasting message is that matching markets can have prices even when money is absent. The prices are cutoffs, and those cutoffs are enough to bring supply-demand reasoning into school choice, college admissions, and related market-design environments.